MRITYUNJAY BALKRISHNAN

Atlanta, Georgia • Email • LinkedIn • Github

EDUCATION

Georgia Tech Scheller School of Business, Atlanta **Aug 2026 – Dec 2027(Expected)**
MS in Quantitative and Computational Finance
Expected Coursework: Stochastic Processes in Finance, Derivative Securities, Market Microstructure, Fixed Income Securities

Jadavpur University, Kolkata **Aug 2018 – May 2022**
Bachelor in Engineering (Hons.) CGPA: 8.6/10
Relevant Coursework: Probability, Linear Algebra, Applied statistics, Optimization, Stochastic Processes, Numerical Methods

WORK EXPERIENCE

Assistant Manager – Quantitative Finance, Deloitte, Hyderabad *Jan 2026– Aug 2026*

- Researched systematic relative-value opportunities across commodity calendars and inter-commodity spreads using macroeconomic, inventory, freight and alternative datasets.
- Built commodity forward-curve construction and calendar-spread valuation libraries for derivatives, supporting pricing, risk analysis, and quantitative research workflows.
- Calibrated SABR, Local Volatility and Heston models to exchange-traded commodity option surfaces using nonlinear least-squares optimization, analyzed smile stability across maturities and volatility regimes.
- Applied cointegration, PCA, GARCH, and factor models to identify cross-market risk premia and regime-dependent relationship

Quant Intern- Trading Research, ProAlpha Capital, Mumbai *Nov 2024 – May 2025*

- Researched short-horizon alpha signals from **Level-II order book** data using queue imbalance, order-flow imbalance, liquidity replenishment, and short-term mean reversion features.
- Built event-driven exchange market-making simulators in Python and C++, incorporating queue-position modeling, exchange latency, transaction costs, and realistic fill logic, supported live deployment of strategies with out-of-sample Sharpe of 1.4
- Improved execution quality by modeling adverse selection, slippage, and market impact, increasing realized edge by **1.2 bps**.
- Designed market-making strategies incorporating inventory risk controls, spread optimization, fill probability estimation.
- Conducted microstructure research on signal persistence and execution quality across varying volatility and liquidity regimes.

Senior Quantitative Analyst, Algoritt, New Delhi *Jul 2025 – Jan 2026*

- Researched systematic volatility strategies using carry, skew, and term-structure across equity index options.
- Built a modular QR library supporting pricing, volatility surface construction, calibration, signal generation and backtesting.
- Developed walk-forward research pipelines evaluating **10M+ option quotes** across multiple expiries.
- Calibrated Heston, SABR models using Levenberg–Marquardt optimization with automated parameter stability diagnostics
- Developed statistical validation framework reducing false-positive alpha discoveries via bootstrap and out-of-sample testing.

Data Scientist, JOGGF, New Delhi *Nov 2023 – Jul 2024*

- Built an LLM-assisted financial intelligence (LangChain, FAISS, GPT-4) pipeline processing 20K+ earnings transcripts and analyst reports; fine-tuned transformer-based models and engineered structured features for downstream predictive modeling.

Design Process Engineer, Reliance Industries Limited, Navi Mumbai *Jun 2022 – Oct 2023*

- **Led 13-member** cross-functional team on retrofit projects; applied quantitative risk identification and mitigation strategies.

SELECTED PROJECTS

Options Market-Making Simulator| GitHub |

- Built an event-driven exchange simulator with realistic limit-order-book matching.
- Benchmarked Avellaneda–Stoikov inventory strategies against reinforcement-learning market makers.
- Built post-trade analytics measuring realized spread, adverse selection, inventory PnL, and execution quality.

Low-Latency Limit Order Book Engine | GitHub |

- Built a cache-aware C++17 limit-order-book matching engine with lock-free order processing and latency benchmarking.
- Developed price-level data structures processing 1M+ events/sec.

Derivatives Pricing Library | GitHub |

- Implemented Black-Scholes, Monte Carlo, and finite-difference pricing engines with Greeks and implied-volatility estimation.

Statistical Arbitrage Research | GitHub |

- Developed cointegration-based statistical arbitrage framework using Kalman filters and rolling Johansen estimation.
- Performed walk-forward validation with realistic transaction-cost and execution modeling.

SKILLS

Programming: Python, C++17 , SQL, Linux/Unix, Git, Docker, AWS
Quantitative Research: Market Microstructure, Statistical Arbitrage, Market Making, Time series Econometrics, Monte Carlo
Derivatives: Black-Scholes, Volatility Surface Calibration, Greeks, SABR, Heston, Local Volatility, Calibration
Machine Learning & AI: PyTorch, Transformers, LoRA, XGBoost, RAG, Agents, TensorFlow, Hyperparameter Optimization

CERTIFICATIONS: Akuna Capital 101 (June, 2026), Executive Program in Algorithmic Trading(QuantInsti 2025), Executive Program in Applied Data Science and Machine Learning (IIT Madras , 2023-2024)

PUBLICATIONS & ACHIEVEMENTS

- **All India Rank 26** in Dr C.R Rao Statistics Olympiad by CR Rao AIMSCS **2016**
- Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship with an **All-India Rank 48** **2018**
- **Founded Finance and Consulting Club (Kolkata),** launched the website; **organized 10+** workshops with industry experts.
- **Published research on** ANN based Heat sink optimization and thermal modelling (ISBN: 978-1638321408) **2021**
- **Co-authored** 2 books (Artificial neural network) based projects on active matter molecular Brownian motion & Heat Sink **2021**